STA2002 - Probability and Statistics II

Fall Term, 2023-24

Midterm Examination

October 29, 2023

INSTRUCTIONS:

1. The time duration is 2 hours.
2. The total mark is 100 points.
3. There are total 8 questions, please check.
4. Please write down your calculation and statements clearly and in detail.
5. Please write down ALL your answers on the ANSWER BOOK.
6. Give answers up to 2 decimal places where appropriate.

- $z_{0.005} = 2.58$, $z_{0.01} = 2.33$, $Z_{0.025} = 1.96$, $Z_{0.05} = 1.645$, $z_{0.1} = 1.28$, where $P(Z \geq z_\alpha) = \alpha$ for any $\alpha \in [0, 1]$ with Z being a standard normal random variable.
- $t_{0.005}(12) = 3.05$, $t_{0.005}(13) = 3.01$, $t_{0.01}(12) = 2.68$, $t_{0.01}(13) = 2.65$, $t_{0.025}(12) = 2.18$, $t_{0.025}(13) = 2.16$, $t_{0.05}(12) = 1.78$, $t_{0.05}(13) = 1.77$, $t_{0.1}(12) = 1.36$, $t_{0.1}(13) = 1.35$, where $P(T \geq t_\alpha(df)) = \alpha$ for any $\alpha \in [0, 1]$ with T being a t random variable with degrees of freedom df.
- $t_{0.025}(24) = 2.064$, $t_{0.025}(26) = 2.055$, $t_{0.05}(24) = 1.712$, $t_{0.05}(26) = 1.701$, where $P(T \geq t_\alpha(df)) = \alpha$ for any $\alpha \in [0, 1]$ with T being a t random variable with degrees of freedom df.
- $f_{0.975}(12, 12) = 0.31$, $f_{0.975}(13, 13) = 0.32$, $f_{0.95}(12, 12) = 0.37$, $f_{0.95}(13, 13) = 0.39$, $f_{0.025}(12, 12) = 3.28$, $f_{0.025}(13, 13) = 3.12$, $f_{0.05}(12, 12) = 2.69$, $f_{0.05}(13, 13) = 2.58$, where $P(F \geq f_\alpha(df_1, df_2)) = \alpha$ for any $\alpha \in [0, 1]$ with F being a F random variable with degrees of freedom df_1 and df_2.

1. [10 points] In the following questions, please answer whether each of these statements is true or false.

   1.1 [2 points] In a density histogram, the area of each rectangle is equal to the frequency of the observations in the corresponding interval.

   FALSE

   1.2 [2 points] The method-of-moment estimators for a given model are usually not unique, and we prefer to use the lowest possible moment to build the estimator.

   TRUE

   1.3 [2 points] Let α and β denote the probabilities of type I and type II errors, respectively. It is possible for us to decrease both α and β simultaneously by increasing the sample size.

   TRUE

   1.4 [2 points] John and Mike are both interested in estimating the average GPA of CUHKSZ students. Mike is kind of lazy and only collected data from 10 students. John is quite diligent and collected data from 100 students. They both compute 95% confidence intervals. Assuming their confidence intervals are both valid, John has a greater probability of capturing the true average GPA in his confidence interval than Mike does.

   FALSE

   1.5 [2 points] If a hypothesis is rejected at the 0.05 level of significance, it is possible that it is not rejected at the 0.01 level.

   TRUE

IMPORTANT: For all the questions below, please show your deductions and reasons in detail.

2. [6 points] Let $X_1, \ldots, X_n$ be independent Poisson random variables with means $\lambda_1, \ldots, \lambda_n$, that is, $X_i \sim Poisson(\lambda_i)$, where $\lambda_i > 0$, for $i = 1, \ldots, n$. Identify the distribution of $\sum_{i=1}^n X_i$ using the moment generating function technique. Note that for a Poisson random variable $X \sim Poisson(\lambda)$, its moment generating function is $e^{\lambda(e^t-1)}$.

   Answer: MGF for $\sum_{i=1}^n X_i$ is $e^{\sum \lambda_i(e^t-1)}$. Hence, $\sum_{i=1}^n X_i$ is $Poisson(\sum \lambda_i)$.

3. (a). [6 points] Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with pdf

$$f(x|\alpha) = \alpha^2 x e^{-\alpha x}$$

   for $X \geq 0$ and $\alpha > 0$. Derive the maximum likelihood estimator for α.

   Answer: The log likelihood is $2n \log \alpha - \alpha \sum x_i + \sum \log x_i$. We solve $2n/\alpha - \sum x_i = 0$, which yields $\alpha = 2n/\sum x_i$.

(b). [6 points] Let X be a discrete random variable with the following probability mass function

$$Pr(X = x|\theta) = \begin{cases} \theta^2 & x = 0 \\ 2\theta(1-\theta) & x = 1 \\ \theta^2 & x = 2 \\ 1 - 2\theta & x = 3 \end{cases}$$

where $0 < \theta < \frac{1}{2}$. If $X_1, \ldots, X_n$ are n i.i.d. observations on on X. Derive a method of moment estimator for θ.

Answer: $EX = 2\theta(1-\theta) + 2\theta^2 + 3 \times (1 - 2\theta) = 3 - 4\theta$. $\hat{\theta} = (3 - \bar{X})/4$.

(c). [6 points] Suppose $E(X) = \mu$ and $Var(X) = \sigma^2$, and X_1 and X_2 are two i.i.d. observations on X. Show that $\frac{1}{2}(X_1 - X_2)^2$ is an unbiased estimator of σ^2.

Answer: $E(X_1 - X_2) = 0$. Thus,
$E\left[\frac{1}{2}(X_1 - X_2)^2\right] = \frac{1}{2}Var(X_1 - X_2) = \frac{1}{2}\left[Var(X_1) + Var(X_2)\right] = \sigma^2$.

4. [12 points] An article in *Computers & Electrical Engineering* considered the speedup of parallel cloud computing. Experiments on the speed are performed and the data are as follows (in seconds):

$$3.7, 3.3, 4.2, 4.0, 4.6, 4.1, 4.3, 4.8, 4.2, 4.6, 4.9, 4.3, 4.6.$$

The sample mean speedup is 4.2769, and the sample standard deviation is 0.4456. Suppose the speeds are normally distributed with unknown mean and variance.

(a). [6 points] Construct the 95% two-sided confidence interval on the mean speedup.

Answer: Use $t_{0.025}(12) = 2.18$. Lower bound is $4.2769 - 2.18 \times 0.4456/\sqrt{13}$; upper bound is $4.2769 + 2.18 \times 0.4456/\sqrt{13}$.

(b). [6 points] Now assume the variance of the speedup is known as 0.5. An engineer wishes to estimate the mean speedup by performing the experiments. At least how many experiments does the engineer need to perform if she wants to be 99% confident that the sample mean is within 0.02 of the true mean speedup?

Answer: Use $z_{0.005} = 2.58$. $n = (2.58/0.02)^2 \times 0.5$. Round it up.

5. [16 points] Charles has instructed Michael, who is an employee at Charles' bakery store, to add on average 10 blueberries to muffins. He would like see if Michael has been following his instruction. To be sure, he has randomly checked 13 muffins that Michael made and counted the numbers of blueberries contained in these muffins. The sample mean and sample standard deviation in turns are 11 and 2. Charles is considerate, and decides that he doesn't want to alarm Michael unless his findings are significant at the $\alpha = 0.01$ level.

(a). [6 points] Write down the hypothesis test being conducted above. Explain what the type I and type II errors are in the context of this problem.

Answer: $H_0 : \mu = 10, H_1 : \mu \neq 10$.

Type I error: say Michael did not follow the instruction (i.e., to add on average 10 blueberries) when he actually did.

Type II error: say Michael followed the instruction (i.e., to add on average 10 blueberries) when he actually did not.

(b). [5 points] Define the rejection region at the $\alpha = 0.01$ level. Does Charles reject H_0, the null hypothesis? Justify your answer.

Answer: Test statistic: $T = \frac{\bar{X}-10}{\sqrt{s^2/n}} \approx 1.8$.

Critical value: $t_{0.005}(12) = 3.05$

Reject region: When $T > 3.05$ or $T < -3.05$, reject H_0.

Fail to reject H_0 because T is not in the rejection region.

(c). [5 points] Explain how you would obtain a p-value? You can just write out the probability statement but don't need to calculate the exact p-value.

Answer: $2 \times P(t(12) > 1.8)$.

6. [12 points] According to a population census in 1986, the percentage of males who are 18 or 19 years old and are married was 3.7%. We would like to see if there is evidence that this percentage decreased from 1986 to 1988.

(a). [2 points] Define the null and alternative hypotheses.

Answer: $H_0 : p = 0.037, H_1 : p < 0.037$.

(b). [5 points] Define a rejection region that has an approximate significance level of $\alpha = 0.05$. If $y = 30$ out of a random sample of $n = 1000$ males, each 18 or 19 years old, were married, what is your conclusion? Justify your answer.

Answer: Test statistic: $Z = \frac{\hat{p}-0.037}{\sqrt{0.037(1-0.037)/1000}} \approx -1.17$.

Critical value: $-z_{0.05} = -1.645$

Reject region: When $Z < -1.645$, reject H_0.

Fail to reject H_0 because Z is not in the rejection region.

(c). [5 points] What is the power for the test based on your rejection region given in (b)? You can just write out the probability statement but don't need to calculate the exact value of the power.

Answer: For any $p_1 < 0.037$. The power is

$$P\left(\frac{\hat{p} - 0.037}{\sqrt{0.037(1-0.037)/1000}} < -1.645 \middle| p = p_1\right).$$

Give the students three points if they can obtain the above expression. If they can further elaborate, e.g., by saying the distribution of $\hat{p}$ or that of the binomial random variable under $p = p_1$, then give them the full points.

7. [12 points] Suppose that $X_1, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$ and σ^2 is known. We would like to test

$$H_0 : \mu = \mu_0, \quad H_1 : \mu < \mu_0.$$

We consider the test statistic

$$T = \overline{X}$$

and the rejection region at the significance level α

$$C = \left\{\overline{x};\ \overline{x} < \mu_0 - z_\alpha \frac{\sigma}{\sqrt{n}}\right\}.$$

(a). [6 points] We write $\beta(\mu')$ to be the probability of type II error given the true value of $\mu = \mu'$. Prove that

$$\beta(\mu') = \Phi\left(z_\alpha - \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right),$$

where $\Phi(z) = P(Z \le z)$ is the cdf of $Z \sim N(0,1)$.

Answer:

$$P\left(\bar{X} > \mu_0 - z_\alpha \frac{\sigma}{\sqrt{n}} \middle| \mu'\right) = P\left(Z > \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}} - z_\alpha\right) = P\left(Z < -\frac{\mu_0 - \mu'}{\sigma/\sqrt{n}} + z_\alpha\right) = \Phi\left(z_\alpha - \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right).$$

(b). [6 points] Consider testing $H_0 : \mu = 10$ versus $H_1 : \mu < 10$ using data from a normal population distribution with $\sigma = 2$. Assuming the true value of μ is $\mu' = 9$. Under the level of significance $\alpha = 0.05$, what is the sample size n needed in order for the power to be at least 90%?

Answer:

$$n = (1.645 + 1.28)^2 \times 2^2/(10-9)^2 = 34.225.$$

Use $n = 35$.

8. [14 points] Two different analytical tests can be used to determine the impurity level in steel alloys. Assume that the impurity level under both tests follow normal distributions, i.e., the impurity level under the first test follows $\mathcal{N}(\mu_1, \sigma_1^2)$, and the second follow $\mathcal{N}(\mu_2, \sigma_2^2)$.

Thirteen specimens (samples) are tested using each of the two procedures, i.e., $n_1 = n_2 = 13$. The sample mean for the first test is $\bar{x}_1 = 1.48$, and for the second test is $\bar{x}_2 = 1.69$. The sample standard deviation for the first test is $s_1 = 0.307$, and for the second test is $s_2 = 0.377$.

(a). [7 points] To compare the mean impurity levels between the two procedures, the statistician would like first to find out whether equal variance between the procedures can be assumed, that is to first conduct a test for the equality of variance. State the hypotheses form clearly and show your reasons. Use the significance level $\alpha = 0.05$.

Answer: $H_0 : \sigma_1^2 = \sigma_2^2, H_1 : \sigma_1^2 \neq \sigma_2^2$.
$F = 0.307^2/0.377^2 \approx 0.66$.
$f_{0.975}(12, 12) = 0.31$. Since $0.66 > 0.31$, don't reject H_0.

(b). [7 points] Based on your conclusion from item (a), is there sufficient evidence to suggest the mean impurity level of the second procedure is significantly higher than that of the first one? Conduct a t-test to compare the group means. Please state the hypotheses form clearly and show your reasons. Use the significance level $\alpha = 0.05$.

Answer: $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 < \mu_2$.
Pooled sample variance: $(12 \times 0.307^2 + 12 \times 0.377^2)/24 = 0.118189$.
Test statistic is

$$\frac{1.48 - 1.69}{\sqrt{0.118189 \times 2/13}} \approx -1.56.$$

Critical value is $-t_{0.05}(24) = -1.712$. Don't reject H_0 since $-1.56 > -1.712$.